ISL Fingerspelling Recognition System

Step-by-step block diagram labels and drawing order

1. Data Collection

- ISL Dataset
- Approximately 14,000 images across 35 classes

2. Data Preprocessing

- Image preprocessing: resize and normalize images
- Landmark extraction: MediaPipe, 42-D landmark features

3. Dataset Splitting

- Training set: 70%
- Testing set: 30%

4. Model Implementation

- CNN (MobileNetV2)
- CNN (EfficientNet-B0)
- Landmark DNN (42-D)
- Hybrid Late Fusion
- Baseline models: CNN-only and Landmark-only

5. Explainability and Guidance

- Grad-CAM heatmaps
- t-SNE feature plots
- Gemini sign guidance
- Applied to the hybrid model

6. Model Performance Evaluation Metrics

- Accuracy, Precision, Recall, and F1 Score (macro/weighted)
- Standard 70/30 split
- Signer-independent evaluation
- Background-independent evaluation

7. Best Model Selection

- Compare evaluation results and select the best-performing model
- Planned candidate: Hybrid Late Fusion (confirm using actual results)

8. Final Model

- Hybrid CNN + Landmark Late Fusion
- 35-class Softmax output

9. Model Outcome

- Generate the predicted class for the input hand sign

10. Final Output

- Recognized sign from the 35 classes
- Gemini sign guidance, where applicable

Main drawing flow

Data Collection → Data Preprocessing → Dataset Splitting → Model Implementation → Model Performance Evaluation → Best Model Selection → Final Model → Model Outcome → Classification → Final Output

Drawing note: Training and testing should be shown as two branches after dataset splitting. Explainability and guidance support interpretation of the model and can be shown alongside evaluation. Only label Hybrid Late Fusion as the best model after confirming the measured results.